

SQL Master Librarian: Managing & Connecting Data

Comprehensive Lesson Guide & Hands-On Coding Activity Worksheet (INSERT, UPDATE, DELETE, JOIN)

COMPUTER SCIENCE & DATABASES | PLANRS ACADEMY

1. Introduction: Real-World Relational Databases

Ever wonder how food delivery apps like Zomato know your order, address, and delivery partner, or how your school keeps track of thousands of students?

They use **Relational Databases** where data is structured into grids called **Tables** (rows and columns).

- **Table:** Collection of records (rows) with attributes (columns).
- **Primary Key:** A unique identifier for every row (e.g., RollNo or StudentID). No two records can share the same Primary Key.

2. Key SQL Data Manipulation Commands (DML)

Example 1: Adding New Records using INSERT INTO

When a new record needs to be added, specify table name, columns, and values:

```
-- Adding a new player to the Players table
INSERT INTO Players (PlayerID, Name, Runs, Team)
VALUES (7, 'Virat', 0, 'India');
```

Step-by-Step Walkthrough:

1. `INSERT INTO Players` targets the table where data will be added.
2. `(PlayerID, Name, Runs, Team)` explicitly lists the column order.
3. `VALUES (7, 'Virat', 0, 'India')` provides exact matching values (numbers without quotes, text in single quotes).

Example 2: Modifying Existing Records using UPDATE

Update specific column values for existing rows using a filter condition:

```
-- Updating Virat's score after a match
UPDATE Players
SET Runs = 75
WHERE PlayerID = 7;
```

Step-by-Step Walkthrough:

1. `UPDATE Players` targets the table to modify.
2. `SET Runs = 75` changes the value in the Runs column.
3. `WHERE PlayerID = 7` ensures only player #7 is modified, preserving all other player records.

Example 3: Removing Records using DELETE

Remove rows that are no longer needed from a database table:

```
-- Removing retired player with PlayerID 3
DELETE FROM Players
WHERE PlayerID = 3;
```

Step-by-Step Walkthrough:

1. `DELETE FROM Players` specifies the table from which data will be deleted.
2. `WHERE PlayerID = 3` targets specifically player #3 for removal.

⚠ CRITICAL SAFETY RULE FOR UPDATE & DELETE:

Always include a `WHERE` clause! Executing `DELETE FROM Players;` or `UPDATE Players SET Runs = 0;` without `WHERE` will permanently wipe or reset all data in the table!

3. Connecting Data Across Tables: SQL JOIN

Applications split data across separate tables to prevent redundancy. We connect them using shared key columns.

Example 4: Linking Tables using INNER JOIN

Suppose we have two related tables: `Students` and `Library` linked by `RollNo`.

```
-- Query: Find student names and titles of borrowed books
SELECT Students.Name, Library.BookTitle, Library.ReturnDate
FROM Students
INNER JOIN Library
    ON Students.RollNo = Library.RollNo
WHERE Library.BookTitle = 'Panchatantra';
```

Step-by-Step Walkthrough:

1. `SELECT Students.Name, ...` picks specific columns from both tables using `TableName.ColumnName`.
2. `FROM Students INNER JOIN Library` connects the two tables.
3. `ON Students.RollNo = Library.RollNo` specifies the matching key columns.
4. `WHERE Library.BookTitle = 'Panchatantra'` filters for a specific book title.

Example 5: Multi-Table JOIN Query (Orders & Customers)

Combining order records with customer details to generate an invoice view:

```
-- Retrieve customer details along with item details for orders
SELECT Orders.OrderID, Customers.CustomerName, Orders.ItemName, Orders.Price
FROM Orders
JOIN Customers
    ON Orders.CustomerID = Customers.CustomerID
WHERE Orders.Price > 100;
```

Step-by-Step Walkthrough:

1. Links `Orders` and `Customers` via common key `CustomerID`.
2. Filters results to show orders where price exceeds 100.

Student Worksheet & Practice Questions

Test your SQL skills as the Master Librarian! Read each scenario carefully and answer.

Section A: Multiple Choice & Conceptual Assessment

Question 1

Scenario: A new student named Priya has newly joined Class 6B. You need to add her information to the school database. Which command should you use?

- (A) UPDATE (B) INSERT (C) DELETE (D) JOIN

Question 2

Scenario: Priya moved to a new address. You need to modify her address in the system without altering other records. Which command will you execute?

- (A) JOIN (B) DELETE (C) UPDATE (D) INSERT

Question 3

Scenario: A library book titled "Panchatantra" was damaged beyond repair and must be removed permanently. Which command performs this task?

- (A) DELETE (B) INSERT (C) UPDATE (D) JOIN

Question 4

Scenario: You have a "Students" table and an "ExamScores" table. You want to generate a report showing student names alongside test marks. Which command links these tables?

- (A) UPDATE (B) JOIN (C) INSERT (D) DELETE

Section B: Step-by-Step Hands-On Coding Exercises

Question 5: Writing INSERT Queries

Write a complete SQL query to insert a new book into the Library table with values: BookID = 501, BookTitle = 'The Secret Garden', and Author = 'Frances Hodgson Burnett'.

Question 6: Writing UPDATE Queries with Safety

Write a step-by-step SQL query to update the price of a book titled 'Database Essentials' to 450 in the Books table. Include a comment explaining why WHERE is necessary.

Question 7: Writing DELETE Statements

Write an SQL statement to delete all records from the Borrowers table where ReturnStatus = 'Returned'. What happens if you omit the WHERE clause?

Question 8: Constructing a Table JOIN

Given two tables: Orders (OrderID, CustomerID, ItemName) and Customers (CustomerID, CustomerName, Address), write an SQL JOIN query to display CustomerName and ItemName for all orders.

Question 9: Advanced Multi-Step Query Challenge

A student named 'Rahul' (RollNo = 102) returns the book 'Harry Potter' (BookID = 305). Write two SQL queries:

- 1) An UPDATE query to set ReturnDate = '2026-08-19' in the Library table for BookID = 305.
- 2) A SELECT JOIN query to display Rahul's Name and the updated ReturnDate.